

AegisStorm AI

A responsive single-page cyclone-risk and vulnerability dashboard prototype for the Bay of Bengal / Odisha coast. Built with Next.js, React, Tailwind CSS, Leaflet / React-Leaflet and Lucide React.

Run locally

Bash

```
npm install
npm run dev
```

Open the local Next.js preview at the URL/port printed by the development server. `npm run lint` runs ESLint; `npm run tsc` checks TypeScript types.

Prototype scope

- Leaflet map centered at 19.8°N, 85.8°E, using OpenStreetMap tiles, interactive hazard circles, a Route 16 corridor and critical-infrastructure popups.
- Scenario selector for Cyclone Fani, Cyclone Amphan and a Category 4 simulation.
- DaisyUI themes are enabled through Tailwind CSS v4, with System, light, dark, cupcake, synthwave, cyberpunk and retro choices in the header; the selected theme persists through `next-themes`.
- The “Gemini Multimodal Vulnerability Assessment” control produces local, illustrative advisory content. It is a frontend simulation and does not call Gemini or any external AI service.
- The SAR inundation-index panel is mock data and is not live Google Earth Engine output.
- The cyclone parameters, modeled exposure, forecast window and alerts in this demo are illustrative. This is not an official warning or operational emergency-management product.

Image attribution

Cyclone Fani satellite image: NOAA National Environmental Satellite, Data, and Information Service (NESDIS), [“Cyclone Fani Batters India's Coastal Odisha State”](#). The selected image is bundled at `public/images/cyclone-fani.webp` for the project preview.